

AeroCAD

AI-Assisted Urban Cadastral Mapping & Verification

Smart India Hackathon 2026

Problem Statement 26012 | SIH 2026 | land records | geospatial context

Problem Statement ID	26012
Problem Statement	AI-Based Automated Urban Parcel Mapping and Cadastral Feature Extraction System using Drone Imagery
SIH organization / department metadata	Project record supplied for this report lists AICTE; current 2026 online SIH listings associate PS 26012 with the Ministry of Rural Development / Department of Land Resources (DoLR). See the metadata note in Section 3.
Project version documented	AeroCAD v0.20.0
Report basis	Project planning/development notes, final code package audit, and current external sources checked 29 September 2026

“AI proposes. GIS validates. Humans verify.”

This report deliberately separates demonstrated implementation from planned work. It does not claim that AeroCAD is legally authoritative, production-ready, or proven to outperform existing cadastral workflows.

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1 Executive Summary

AeroCAD began with a broad problem statement about urban cadastral mapping, but the real difficulty only became clear after working through the data and the workflow. The problem is not simply drawing parcel polygons from drone images. A parcel boundary may not be visible in the image at all, while buildings, roads, terrain, existing cadastral records and field observations can all provide pieces of evidence. The project therefore evolved into a platform that combines geospatial preprocessing, feature extraction, candidate parcel generation, topology checking, confidence indicators, field verification and GIS export.

The current AeroCAD prototype can accept ORI/orthomosaic and DSM inputs, with optional DTM, existing parcel GIS, ground-truth and GNSS data. It produces candidate building and road features, candidate parcels, quality-control outputs, a field-review queue and GIS-ready exports. Real Telangana data was used during development, including work centred on Angadi Kistapur and Yavapur; the Telangana Remote Sensing Applications Centre (TGRAC) itself documents drone-based Abadi mapping pilot work in those two locations. [R1]

At the same time, this report does not treat every screen in the interface as proof that every underlying capability is finished. The project record contains a clear set of remaining technical and validation tasks, and the final packaged repository still contains some legacy frontend demo arrays. The honest position is therefore that AeroCAD is a substantial end-to-end prototype whose core workflow is implemented, but whose specialist model validation, final spatial calibration, complete audit chain, and several operational refinements still require further work.

2 Where the Story Starts

A parcel map begins with a line. On a screen, that line can look simple. In a land-record system, however, that line has consequences. It can represent the edge of a property, a change in land use, an access route, or a place where a surveyor needs to stop and verify what the image is suggesting.

That is the starting point of AeroCAD. The project was not framed as “build a better map viewer.” It was framed around a much more practical question: how much of the repetitive interpretation and checking around urban cadastral mapping can be supported by software, while still keeping human survey judgment where it is genuinely needed?

The first reading of the SIH statement suggested a fairly direct pipeline: imagery goes in, AI extracts features, parcel polygons come out. The development process quickly showed why that is too simplistic. Buildings are visible, roads are often visible, but legal parcel boundaries are not necessarily visible. That led to the most important design decision in the project: AeroCAD would generate preliminary candidate parcels from multiple sources of evidence and expose uncertainty instead of hiding it.

3 The SIH Problem

The problem statement reproduced in the project record identifies several connected issues: manual interpretation of drone imagery, field-based ground truthing, extraction of parcel boundaries, building footprints and road networks, irregular urban settlement patterns, encroachments, overlapping structures, narrow access roads and delays caused by manual digitization and validation. [P1]

It asks for a system that can use high-resolution drone imagery, orthorectified imagery (ORI), DSM/DTM datasets, existing GIS parcel layers, ground-truth datasets and GNSS/CORS-enabled survey data. It also explicitly calls for AI-based image segmentation, deep learning for feature extraction and object detection, automated topology generation and parcel polygon creation, detection of overlapping or inconsistent parcel geometries, and a Web-GIS visualization and editing interface. [P1]

One metadata point needs to be kept explicit. The problem-statement table reproduced in the supplied project record names AICTE. Current online SIH 2026 listings for PS 26012 associate the problem with the Ministry of Rural Development and the Department of Land Resources (DoLR), and list it as a software problem under Smart Automation. [R5] Because the directly retrievable official SIH owner page was not available during this report check, the report keeps the supplied project record as the project-history source and uses the current online listing only as a metadata cross-check. This avoids turning a source discrepancy into a false certainty.

The expected outcome in the source record is not merely an image segmentation mask. It is an AI-enabled automated cadastral mapping platform that can generate preliminary urban parcel maps, reduce manual effort, improve the speed and efficiency of surveys, support accuracy improvement and support ground-truth and field-verification activities. [P1]

4 Why This Problem Is Hard

4.1 A visible feature is not the same thing as a legal boundary.

Aerial imagery can show a roof, road, wall, fence or terrain break without showing the legal parcel line. Treating the visible image as if it contains a complete legal boundary is therefore too strong a claim for a prototype.

4.2 Urban geometry is messy.

Dense settlements can contain irregular parcels, overlapping structures, narrow passages, small slivers, gaps and invalid source polygons. Even when feature extraction works, the resulting GIS geometry can still need cleanup and review.

4.3 Different data layers may disagree.

ORI, DSM, DTM, existing parcel GIS, ground truth and GNSS observations can have different coordinate systems, resolutions, extents and quality. Bringing them together requires explicit geospatial handling rather than simple overlay.

4.4 Automation creates a second problem: trust.

A system that gives a parcel boundary without showing evidence or uncertainty can create false confidence. AeroCAD therefore treats confidence as a review signal and keeps field verification in the loop.

5 Wider Context: India, Telangana and Recent Developments

AeroCAD is not being developed in isolation. Indian land administration is already moving toward larger-scale digital spatial records, aerial surveying and integrated geospatial workflows. The following developments are included as context, not as evidence that AeroCAD is superior to any existing system.

Date / Source	Source	What was reported	Why it matters to AeroCAD
January 1, 2026	Department of Land Resources / PIB	The Department's 2025 year-end review reported that the NAKSHA urban-land-record pilot had progressed in 157 Urban Local Bodies; aerial flying was completed in 116 ULBs, covering about 5,915 sq. km, while ground truthing had begun in 72 ULBs. The same review reported 97.14% national digitization of cadastral maps.	Shows that urban land-record modernization is already a large-scale operational direction in India. [R2]
Current TGRAC programme information	Telangana Remote Sensing Applications Centre	TGRAC describes work in rural and urban land mapping and cadastral surveys. Its project page specifically documents drone-based Abadi mapping pilots in Angadi Kistapur (Siddipet) and Yavapur (Medak), including generation of high-resolution maps and building information.	This is directly relevant to the project's choice of Telangana data and study areas. [R1]

September 22, 2026	The Times of India	MCD reported a pilot drone-based GIS and 3D digital survey in parts of Delhi for property-tax assessment, using drone imaging, LiDAR, GIS, AI/ML, geo-tagging and 3D modelling. The stated aim was to identify mapping/property-record discrepancies and reduce reliance on manual inspections.	A current real-world example of municipalities using aerial and geospatial technologies for property information workflows. [R3]
SVAMITVA programme	Ministry of Panchayati Raj	SVAMITVA uses drone mapping in rural inhabited (Abadi) areas and has built a large national workflow around spatial mapping and property cards.	Useful background because it demonstrates that drone-based property mapping is already being used in a formal Indian land-administration programme; it should not be confused with the urban NAKSHA programme or with AeroCAD itself. [R4]

The project uses these developments as context only. It does not claim that AeroCAD is already equivalent to a government land-record programme, nor that any external initiative validates AeroCAD's performance.

6 How the Project Chose Its Ground

The team initially searched for freely downloadable Telangana datasets that could provide a common geography for imagery and parcel information. The development record notes that some government routes did not provide a simple public download path and that Survey of India products were not suitable for this short prototype effort because access involved paid ordering. The project therefore moved toward sources that could be used for a documented prototype without purchasing survey products.

This led to Telangana-specific work using TGRAC data and, importantly, to locations with an official connection to drone-based Abadi mapping. TGRAC's own project description names Angadi Kistapur GP in Siddipet District and Yavapur GP in Medak District as pilot-study locations. [R1]

Two real-data project contexts emerged during development: a Kistapur project with 216 parcel candidates and a Yavapur project with 491 parcel candidates. In the Yavapur processing run documented in the project record, the system reported 161 building candidates and 49 road

candidates. Those values are treated here as outputs of the prototype run, not as independently verified ground-truth counts. [P2]

7 From Idea to Working Platform

The build did not happen in one pass. The development record shows a progression from a visual dashboard prototype into a real geospatial backend and then into workflow hardening. Each stage exposed a different class of problem.

Stage	What changed	Why it mattered
Early foundation	AeroCAD skeleton	Established the dashboard, project concepts, data-ingestion screens and a consistent visual language.
v0.3	GeoAI workflow	Added the seven-stage processing story, cadastral map, topology QC and field-verification screens. The UI still used deterministic preview geometry at this point.
v0.5	Real geospatial backend	Introduced real ORI processing, tile-based raster handling, DSM alignment, configurable semantic-segmentation integration, vectorization and explicit fallback behavior.
v0.11.x	Editing / reprocessing	Added real parcel editing, coordinate-oriented boundary handling, rerun processing, topology repair for invalid source geometries and stronger QC logic.
v0.14-v0.16	Persistence / workflow	Moved from isolated testing toward persisted projects, project context, field verification and a more continuous workflow.
v0.17-v0.19	Officer-facing product	Reworked the overview, projects register, reports, settings and prototype account access while keeping the interface intentionally restrained rather than visually overdesigned.
v0.20.0	Repository consolidation	Cleaned versioning and runtime packaging, hardened several backend and GIS paths, and prepared a small GitHub-oriented repository. This version is the report's reference build.

One lesson runs through the entire history: every time a feature looked correct at the UI level, the next question was whether the underlying project state, geometry or export was also correct. That is why the project eventually shifted from feature accumulation to consistency and verification.

8 What AeroCAD Built

Data ingestion and preflight

Accepts ORI, DSM and optional DTM, existing parcel GIS, GT and GNSS/CORS data. Reads CRS, raster size, resolution/GSD information and integrity details before processing.

Tile-based aerial processing

Processes aerial imagery in manageable windows and creates building and road candidate outputs.

Boundary evidence

Combines imagery, structures, access, terrain and existing GIS context rather than relying on a single black-box parcel detector.

Candidate parcel inference

Generates parcel polygons with IDs, area, land-use fields, confidence/evidence information and workflow state.

Topology QC

Checks the generated geometry for overlaps, gaps, slivers and invalid shapes, with parcel-level review.

Field verification

Provides a queue for uncertain parcels, review outcomes, notes and optional GNSS observations.

GIS editing

Supports boundary editing and parcel split/merge workflows followed by revalidation.

Reports and export

Produces GIS-oriented outputs such as GeoJSON, GeoPackage and Shapefile packages together with supporting records.

Project persistence

Persists named survey workspaces on the backend rather than relying only on a hard-coded frontend demo.

9 Technical Logic

AeroCAD is deliberately split into several layers so that an evaluator or future developer can see where a result came from.

INPUT DATA

|

v

DATASET VALIDATION

|

v

GEO-PREPROCESSING

```
|  
+-- Building extraction  
+-- Road / access extraction  
+-- Land-use / contextual evidence
```

```
|  
v
```

BOUNDARY EVIDENCE FUSION

```
|  
v
```

CANDIDATE PARCEL INFERENCE

```
|  
v
```

POLYGON + TOPOLOGY QC

```
|  
+-- confidence / evidence  
+-- geometry issues  
+-- review priority  
|  
v
```

HUMAN GIS REVIEW

```
|  
+-- edit  
+-- split / merge  
+-- revalidate  
|  
v
```

FIELD VERIFICATION

```
|  
v
```

GIS-READY OUTPUT + AUDIT DATA

The current backend uses Python/FastAPI together with Rasterio, GeoPandas, Fiona, Shapely, PyProj and NumPy, with optional specialist model paths through PyTorch, Transformers, ONNX Runtime and Ultralytics. The frontend uses React and Vite. The packaged local prototype keeps project artifacts on disk instead of requiring PostGIS or GeoServer for a basic run.

A particularly important implementation choice is the fallback path. The specialist building and road models are lazy-loaded. If a specialist cannot load or returns no usable mask, the system can use deterministic baseline extraction and record that condition. This avoids a misleading situation where the interface says “AI model ran” while the output actually came from a fallback.

10 Testing and What the Results Actually Mean

Testing happened at two levels: real-data development runs and controlled synthetic smoke tests.

10.1 Real-data development runs

The development record documents successful work with Telangana-specific imagery and parcel data, including Kistapur and Yavapur projects. It also records a real Yavapur source-data failure caused by invalid polygon geometries. The source geometry was repaired only for analytical operations so processing could continue, while the invalid source condition remained a QC concern. [P3]

10.2 Synthetic end-to-end validation

The final v0.20 packaging work was tested with a synthetic survey workflow through validation, ORI preview, feature extraction, parcel inference, evidence fusion, topology, field-verification recording, geometry editing/revalidation, reprocessing and GIS export. This is evidence that the backend/software path can execute end-to-end; it is not a field-accuracy benchmark.

10.3 Frontend build limitation

A complete Vite production build was not completed in the report-generation environment because the pinned Vite package was unavailable in the local npm cache and repeated dependency-install attempts timed out. This is reported as an environment limitation; it is not being presented as evidence that the frontend source is production-build verified.

10.4 What the reported counts mean

Values such as 491 parcels, 161 building candidates and 49 road candidates are outputs produced by the prototype workflow. They should not be written in a report as “ground truth,” “correct counts,” or “accuracy” without a separate labelled evaluation.

10.5 What is still missing for a stronger evaluation

A defensible performance study would need a labelled validation set, explicit metrics for buildings/roads/land use/parcel boundaries, a defined baseline, multiple test areas and a transparent way to measure spatial deviation and topology quality. The project record itself explicitly warned against inventing numerical accuracy claims.

11 The Unfinished Work: All Remaining Roadmap Items

There is an important numbering discrepancy in the source conversation. At one point the work was described as “11 unfinished points,” but the final written roadmap actually contains 12 numbered items. For accuracy, this report records all 12 rather than silently dropping the last item. [P4]

o.	Item	Status	What remains
1	Final GIS / map calibration	Not fully complete / not fully validated	Real raster-to-map transformation, overlay alignment, layer visibility, parcel clicking, zoom/pan, coordinate conversion and vertex edit behavior still needed a dedicated calibration pass.
2	Specialist building + road AI verification	Not fully complete	Specialist model integration exists in code, but the project record did not establish that final Telangana outputs were actually produced by the specialists rather than fallback CV. No specialist accuracy claim is made.
3	Land-use evidence validation	Partially implemented	The workflow can assign land-use information, but the project record says this was not yet one of the strongest demonstrated capabilities and needed clearer evidence and validation.
4	Global active-project context audit	Partially addressed	A central project context was added, but older frontend paths still referenced localStorage directly. A fresh cross-screen audit was still needed to ensure Dashboard, Map, AI Processing, QC, Field Verification and Reports always use the same project.
5	Invalid-source geometry QC verification	Partially addressed	Processing can repair invalid source geometry for analysis, but the desired behavior is to keep the invalid source identifiable and visible in QC instead of silently turning it into a “perfect” dataset.
6	GIS export inspection	Partially addressed	Export paths were implemented and hardened, but the project record still called for inspection of an actual final package for CRS, geometry validity, attributes, duplicates, temporary fields and privacy-sensitive source attributes.
7	End-to-end audit trail test	Partially implemented	Edit history and field-verification records exist, but the complete chain from processing -> QC -> field decision -> edit -> revalidation -> export had not been demonstrated as one continuous audit sequence.
8	Field queue filters	Enhancement remaining	Priority sorting exists, but the roadmap calls for practical filters/search such as Critical/High/Medium/All, parcel search, survey-number search, encroachment and topology flags.
9	Project archive	Not fully implemented	The roadmap preferred Archive instead of hard delete for survey records. The packaged prototype still exposes a delete route, so archive behavior should be treated as unfinished rather than assumed complete.
10	Login / security	Prototype-only	The current login is a local browser-session prototype. It is not production identity management, database authentication, role-based authorization or government SSO.

11	Notifications	Prototype-only	The bell is functional as a project-context popover, but it is not yet an event engine for live survey alerts such as completed processing, critical parcels or topology changes.
12	Remove old demo code	Still present in packaged source	The final frontend source still contains legacy demo parcel arrays and a static “Hyderabad North – Ward 14” dashboard card. These are not the live backend project path, but they should be removed before a final public release to avoid accidental fallback or confusion.

12 What We Can Claim – and What We Cannot

Evidence level	Statement	Meaning
Supported by the project record	AeroCAD implements an end-to-end prototype workflow for geospatial ingestion, processing, parcel inference, topology/QC, human review and GIS export.	The system is working software, not only a presentation mock-up.
Supported by the project record	The prototype was developed and tested with Telangana-specific data, including Kistapur and Yavapur contexts. [R1]	The choice of study area is grounded in real source material, including TGRAC’s documented pilot locations.
Supported by the project record	The software has deterministic fallbacks when specialist inference is unavailable.	The fallback is explicit rather than hidden.
Supported by the project record	The prototype produces candidate parcel, building and road outputs.	“Candidate” is the correct term; these are not automatically ground-truth labels.
Not demonstrated	AeroCAD is more accurate than manual cadastral mapping.	No controlled benchmark was completed, so the report does not make this claim.
Not demonstrated	The specialist building/road models have known final accuracy on the Telangana study areas.	The project record says that specialist verification remained to be completed.
Not demonstrated	AeroCAD determines legal property ownership or confirms encroachment.	The prototype only produces spatial candidates and review flags; legal determination remains outside scope.
Not demonstrated	The prototype is ready for nationwide or production government deployment.	Authentication, CORS integration, scale, model validation, spatial calibration and operational governance remain open areas.

13 Limitations

Model validation

The project has specialist model integration, but not a complete measured evaluation of the specialists on the final Telangana study areas.

Spatial calibration

The map/geometry layer remains the largest technical risk in the project because visual cor-

rectness depends on consistent raster extents, CRS transformations and screen-coordinate mapping.

Ground truth

The project uses reference GIS and survey-oriented inputs, but does not have a sufficiently broad independently labelled benchmark to support a formal accuracy claim.

Legal interpretation

The prototype produces preliminary cadastral information. It does not decide ownership, adjudicate boundary disputes or establish legal encroachment.

Operational security

Login is a prototype local session, not a production access-control system.

Notifications and archive

The operational features around event notifications and survey-record lifecycle are still basic.

Legacy frontend content

Static demo arrays remain in the source and should be removed in the final public release.

Scale

The local prototype uses disk-based project artifacts. A production system would need stronger storage, concurrency controls, monitoring and spatial infrastructure.

14 Future Development

The next version should not be defined by adding more screens. The highest-value work is the work that reduces uncertainty about whether the current outputs are spatially correct and scientifically defensible.

1. Complete the dedicated GIS/map calibration using one known raster extent, one known parcel and one controlled coordinate transformation test.
2. Run and document the specialist building and road models on the Telangana study imagery, and compare their outputs with a reference layer before reporting any model metrics.
3. Strengthen land-use evidence so every classification can be explained in terms of observable spatial features.
4. Finish the global active-project audit so every screen and API operation uses one project source of truth.
5. Expose invalid source geometries as QC findings rather than allowing repaired analysis geometries to create a misleading 100% health result.
6. Inspect the final GIS package in a desktop GIS and confirm CRS, geometry validity, fields and absence of unnecessary sensitive attributes.
7. Complete one end-to-end audit trail from processing through export.

8. Remove the legacy frontend demo data before the public release.
9. Keep production authentication, PostGIS, CORS and large-scale deployment as a later deployment track rather than mixing them into the prototype core.

15 Conclusion

AeroCAD started as a response to a difficult problem statement and became a much clearer engineering problem through implementation. The important change was not simply adding more AI. It was learning what a cadastral workflow actually needs: good input data, spatial consistency, candidate geometry, topology checks, evidence, human review and a usable output.

The prototype now has a coherent path from geospatial input to a reviewable cadastral result. That is a meaningful piece of work in itself. At the same time, the remaining roadmap matters because cadastral systems punish false confidence. A visually convincing map is not enough; the coordinates must line up, the source geometry must be traceable, the model output must be measured, and the human reviewer must know what still needs to be checked.

For that reason, the most responsible description of AeroCAD at v0.20.0 is:

“An AI-assisted urban cadastral mapping and verification prototype that converts drone and geospatial inputs into preliminary parcel candidates, spatial quality checks and human-review workflows.”

The project has moved well beyond a static website. Its next step is not to claim that the problem has already been solved; it is to measure, calibrate and validate the parts that are not yet strong enough to support that statement.

References

- R1** Telangana Remote Sensing Applications Centre (TGRAC), Survey of Lands projects. Official source. <https://www.tgrac.telangana.gov.in/trac/surveyoflands-project.php>
- R2** Press Information Bureau, Department of Land Resources, Year-End Review 2025, published 1 January 2026. Official source. <https://www.pib.gov.in/Pressreleaseshare.aspx?PRID=2210412&lang=2®=48>
- R3** The Times of India, “MCD launches drone survey to plug property tax gaps,” 22 September 2026. Article. <https://timesofindia.indiatimes.com/city/delhi/mcd-launches-drone-survey-to-plug-property-tax-gaps/articleshow/134418822.cms>
- R4** Ministry of Panchayati Raj, SVAMITVA Scheme. Official source. <https://panchayat.gov.in/en/svamitva/>

R5 Current SIH 2026 indexed listing for PS 26012, used only to cross-check the organization/department metadata; the listing associates PS 26012 with the Ministry of Rural Development / Department of Land Resources (DoLR). Cross-check source. <https://sih2026-ps-viewer.vercel.app/ps/SIH26012>